

A Persistence-Weighted Boundary Hierarchy Score for Hi-C Topologically Associating Domains

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Abstract

Topologically associating domains (TADs) are conventionally called by thresholding a one-dimensional boundary-strength signal, most commonly the insulation score, at a fixed cutoff. This treats every called boundary as equally reliable and treats TADs as flat, non-nested segments, even though nested and hierarchical domain structure is well documented in high-resolution Hi-C data. We propose a two-stage pipeline that keeps the standard insulation-score breakpoint step as a fast candidate-boundary filter, then computes persistent homology within the sub-matrix bounded by each pair of consecutive candidate boundaries, across multiple insulation window scales. The resulting persistence value replaces the raw insulation-score magnitude as a boundary confidence score, and nested high-persistence features within a candidate domain surface sub-TAD hierarchy that a flat threshold cannot represent by construction. We state this as two falsifiable claims – that persistence-ranked boundaries better predict CTCF/cohesin ChIP-seq enrichment and cross-replicate reproducibility than insulation-score-ranked boundaries, and that persistence recovers nested domain structure at loci with documented hierarchical organization – and describe the benchmark design, datasets, and compute plan needed to test them. In addition to this empirical comparison, we derive a depth-calibrated confidence bound directly from the persistence stability theorem: combining a Poisson/negative-binomial concentration bound on Hi-C count noise with the bottleneck-distance stability guarantee yields an explicit threshold, as a function of local sequencing depth, above which a candidate boundary’s persistence guarantees its survival under the estimated noise level at a stated confidence. This is the one piece of the pipeline with a provable guarantee rather than an empirical comparison alone; insulation-score magnitude has no analogous property. We do not claim a new standalone TAD caller; we position this work as a confidence and hierarchy layer applicable on top of any existing breakpoint-calling method.

1 Introduction

Hi-C contact matrices reveal that mammalian chromosomes are organized into topologically associating domains (TADs): regions of elevated self-interaction separated by boundaries of reduced interaction [1]. TADs are believed to constrain enhancer–promoter contact and to shape transcriptional regulation, though the precise causal role of TAD boundaries in gene expression remains actively debated.

The dominant computational approach to calling TAD boundaries is the insulation score [2]: a sliding window is passed along the matrix diagonal, contacts within the window are aggregated,

and local minima of the resulting profile are called as boundaries after a strength threshold is applied. Variants of this idea – TopDom, HiCDB, deDoc, and the directionality index – differ in how the local window statistic is computed, but share the same structure: a boundary either clears a threshold or it does not, and the domains between boundaries are treated as flat, non-overlapping units [3].

This is a real limitation for two reasons. First, boundary strength as measured by these methods is a heuristic magnitude with no formal notion of statistical robustness attached to it; the choice of threshold is arbitrary and studies routinely report boundary calls that are sensitive to it. Second, high-resolution Hi-C data show hierarchical, nested domain structure – sub-TADs within TADs – that a single flat threshold on a single scale cannot recover, regardless of how the threshold is chosen.

Topological data analysis, and persistent homology in particular, is a natural candidate for addressing both limitations, because persistence gives a principled, scale-independent measure of how robust a topological feature is, and naturally represents nested structure through the filtration itself [4]. A recent method, topoHiC, has already shown that persistent homology can identify TADs and chromatin loops simultaneously from a Hi-C contact matrix, benchmarked against four loop callers and six TAD callers [4]. That result establishes that whole-matrix persistent homology is a viable approach to domain and loop calling; it does not establish whether a lighter-weight, breakpoint-first design offers a practical advantage over the standard boundary-calling pipelines that the field already runs at scale.

A further, more direct motivation for this project concerns resolution. Insulation score and its variants depend on aggregating counts within a sliding window along the diagonal, which is a reasonable estimator when each window contains enough reads to give a stable average. At ultra-high resolution, 500 bp or 200 bp bins, Hi-C contact matrices are extremely sparse even in the deepest currently available datasets, and there is a genuine open question, raised directly in project discussions with Dr. Muhammad Shamim, of whether insulation-score-based breakpoint calling remains reliable at all once bins are this small. This motivates a third, distinct role for persistent homology in this project beyond ranking and hierarchy: not merely refining insulation-score-derived candidate boundaries, but potentially replacing insulation score as the primary breakpoint detector once resolution is high enough that the windowed-count aggregate it depends on becomes unreliable. We treat this as an explicit, separately falsifiable claim, stated in Section 3, rather than an assumption baked into the pipeline design.

1.1 Contribution and positioning

We do not propose a new whole-matrix TAD caller, and we do not claim to outperform topoHiC’s simultaneous TAD/loop identification. The contribution here is narrower and, we argue, more directly useful: a persistence-based refinement layer that sits on top of an existing, already-standard breakpoint caller. Concretely:

- (i) Stage 1 uses the standard insulation-score method [2] to obtain fast, low-cost candidate boundaries – unchanged from current practice.
- (ii) Stage 2 computes persistent homology only within the sub-matrix bounded by each pair of consecutive candidate boundaries, at multiple insulation window scales, rather than on the whole chromosome. This keeps the computational cost tractable at high resolution and turns boundary confidence into a formal persistence value rather than a heuristic magnitude.

- (iii) Nested high-persistence sub-features within a candidate domain are interpreted as sub-TAD hierarchy, giving a multiscale domain structure that a single insulation-score threshold cannot produce.
- (iv) Boundary confidence is given a provable guarantee, not just an empirical ranking: Section 5 derives an explicit, sequencing-depth-calibrated persistence threshold above which a boundary is guaranteed to survive a stated level of count noise at a stated confidence, using the persistence stability theorem. This is the one component of the pipeline whose justification is a proof rather than a benchmark result.
- (v) At ultra-high resolution (500 bp, 200 bp), where sparsity is expected to degrade insulation-score-based breakpoint calling directly, Section 4.3 describes a persistence-based alternative that does not depend on the same windowed-count aggregate, tested as a distinct, separately falsifiable claim rather than assumed to work by extension of the lower-resolution design.

We state the value of this addition as two falsifiable claims, described in Section 3, and describe the benchmark needed to test them honestly, including the proxy metrics available in the absence of a ground-truth TAD annotation.

2 Related work

Boundary-calling methods. The insulation score [2] remains the most widely used boundary-calling approach and is implemented in `cooltools`, FAN-C, and GENOVA. TopDom, the directionality index, and HiCseg formulate boundary or domain calling as a related but distinct one-dimensional or segmentation problem, generally validated on comparatively low-resolution Hi-C data [3]. HiCDB was proposed specifically to improve sensitivity and robustness of contact domain boundary detection over these earlier scores [3].

Topological methods. topoHiC computes persistent homology over the whole Hi-C contact matrix to identify TADs and loops simultaneously, and reports improved performance over four loop callers and six TAD callers in benchmarking analyses [4]. At the time of writing, no public code implementation of topoHiC is available, so direct, reproducible head-to-head comparison against it is not possible; we rely on its reported benchmark results as an indirect point of comparison and state this limitation explicitly rather than implying a direct replication.

Gap addressed here. No existing method, to our knowledge, combines a fast breakpoint-first step with a windowed, per-domain persistent homology refinement whose explicit purpose is (a) to convert boundary strength into a formal robustness score benchmarked directly against the current standard (insulation-score magnitude), and (b) to do so at a computational cost that scales with the number of candidate domains rather than with the size of the full contact matrix.

3 Falsifiable claims

We commit to two specific, testable claims rather than a general assertion that persistent homology “works better” for TAD calling.

Claim 1 (ranking). Ranking candidate boundaries by persistence value predicts (a) CTCF and cohesin (RAD21/SMC3) ChIP-seq peak enrichment and (b) cross-biological-replicate boundary reproducibility at least as well as, and we hypothesize better than, ranking the same candidate boundaries by raw insulation-score magnitude – the metric current callers already threshold on.

Claim 2 (hierarchy). At loci with documented nested sub-TAD structure, persistence recovers multiple, nested high-confidence boundaries within a single insulation-score-defined domain, which a flat insulation-score threshold cannot represent by construction, regardless of where the threshold is set.

Both claims have a clean negative outcome that would still be reportable: if persistence does not outperform insulation-score magnitude on Claim 1, that is a genuine, useful null result establishing that the added computational machinery is not justified for ranking purposes alone, and the project’s contribution would then rest on Claim 2 in isolation.

Claim 3 (ultra-high-resolution alternative). At 500 bp and 200 bp resolution, a persistence-based breakpoint detector that does not depend on windowed-count aggregation (Section 4.3) recovers boundaries, validated against the same CTCF/cohesin and reproducibility proxy metrics used for Claims 1 and 2, at resolutions where insulation-score-based calling is expected to degrade due to matrix sparsity. This claim is stated separately because it concerns a regime where the Stage 1 baseline itself may not function, rather than a comparison between two working methods; a negative result here, meaning neither approach recovers reliable boundaries at these resolutions, is also a genuine and reportable finding about the limits of current Hi-C data density at this resolution, rather than a failure of the project.

4 Methods

4.1 Stage 1: candidate boundary detection

For a given Hi-C contact matrix at resolution r , we compute the insulation score by sliding a square window of size w along the matrix diagonal and averaging contact values within the window, following [2]. Local minima of the resulting insulation profile, identified via the delta-vector zero-crossing method, are retained as candidate boundaries. This step is run at multiple window sizes $w \in \{w_1, \dots, w_k\}$ to obtain candidate boundaries at multiple scales, matching current best practice in `cooltools`.

4.2 Stage 2: windowed persistent homology

For each pair of consecutive candidate boundaries (b_i, b_{i+1}) at a given scale, we extract the sub-matrix M_i corresponding to that candidate domain, together with a symmetric flanking margin. We construct a filtration on M_i using one of two representations, to be compared empirically before committing to one:

- (a) A Vietoris–Rips complex on a distance matrix $D_i = f(M_i)$, where f maps contact frequency to distance (e.g. $D_i = 1/M_i$ with an appropriate regularization for zero entries), computed with Ripser.
- (b) A cubical sublevel-set filtration computed directly on M_i treated as a 2D image, computed with GUDHI’s cubical complex module.

We compute H_0 and H_1 persistence diagrams for each candidate domain. The persistence of the dominant feature associated with the domain boundary becomes the boundary’s confidence score, replacing the raw insulation-score magnitude in downstream ranking. Nested high-persistence features appearing within a single candidate domain, at finer window scales, are recorded as candidate sub-TAD structure.

Running persistent homology within candidate domains rather than on the full chromosome is a deliberate design choice: it keeps the input size to each persistent homology computation bounded by domain size rather than chromosome size, which is the practical bottleneck for applying topological methods at high resolution (10 kb or finer) genome-wide.

4.3 An alternative breakpoint detector for ultra-high resolution

The two-stage design in Sections 4.1 and 4.2 assumes that Stage 1 supplies usable candidate boundaries for Stage 2 to refine. At 500 bp and 200 bp resolution this assumption itself needs testing, since insulation score depends on a windowed count aggregate that becomes unstable once the average number of reads per bin pair within the window is small, which is expected at these bin sizes even in the deepest currently available Hi-C datasets.

We test a variant of the pipeline in which persistent homology is applied directly to a locally aggregated region of the contact matrix as the primary breakpoint signal, rather than as a refinement of insulation-score-derived candidates. Two concrete adjustments distinguish this from the Stage 2 procedure at standard resolution: first, the region is chosen by a fixed-size sliding scan across the chromosome rather than by insulation-score minima, since insulation-score minima are the object under question at this resolution; second, sparse matrix representations are used throughout, since the contact matrix at 200 bp resolution for a full chromosome has a very large number of bins despite most bin pairs having zero or near-zero counts, and both Ripser and GUDHI’s cubical complex module accept sparse input directly.

Whether the resulting persistence-derived boundaries are more stable and more biologically supported (by the same CTCF/cohesin and reproducibility metrics used elsewhere in this project) than anything insulation score produces at the same resolution is the empirical question in Claim 3. We do not assume in advance that this succeeds; matrix sparsity at 200 bp resolution may be severe enough that neither approach recovers reliable boundaries, and we treat that as a genuine possible outcome to report rather than a failure to avoid.

4.4 Datasets

We use publicly available, already-processed Hi-C contact matrices rather than generating new sequencing data. Primary datasets: Rao et al. GM12878 and IMR90 (GEO GSE63525), accessed as `.cool/.mcool` files via the 4D Nucleome data portal or Juicebox. Where a developmental or comparative-structural angle is useful, Bonev et al. mouse embryonic stem cell differentiation data is included as a secondary dataset. CTCF and cohesin (RAD21, SMC3) ChIP-seq peak calls for the matching cell lines are drawn from public ENCODE tracks.

4.5 Benchmark design

No ground-truth TAD annotation exists; any benchmark claiming to measure “accuracy” against a true TAD set would be overstating what is actually known about TAD structure. We instead use

established proxy metrics:

- **CTCF/cohesin enrichment:** enrichment of CTCF and cohesin ChIP-seq peaks at called boundaries, compared between persistence-ranked and insulation-score-ranked boundary sets at matched call counts.
- **Reproducibility:** Jaccard overlap or stratum-adjusted correlation of boundary calls across biological replicates of the same cell line.
- **Hierarchical recovery:** qualitative and quantitative comparison of nested boundary calls at a small set of loci with documented sub-TAD structure in the literature, against the flat calls produced by insulation score alone.
- **Empirical noise calibration:** bottleneck distance between persistence diagrams computed from matched regions across biological replicates, used both to calibrate ε directly (Section 5) and as a distribution-free cross-check on the parametric Poisson/negative-binomial bound.

Where feasible we report topoHiC’s published benchmark numbers alongside our own for indirect context, flagging explicitly that this is not a reproduced head-to-head comparison, since no public topoHiC implementation is available at the time of writing.

5 A depth-calibrated confidence bound from persistence stability

The comparison in Section 3 treats persistence as a better-behaved ranking signal than insulation-score magnitude, established empirically. This section adds a distinct, provable statement: a threshold, calibrated to the actual sequencing depth of the region in question, above which a candidate boundary’s persistence value guarantees that the boundary survives a stated level of measurement noise at a stated confidence. Insulation-score magnitude has no analogous property, since it is a heuristic statistic with no stability guarantee attached to it; this section is the one part of the pipeline whose justification is a proof rather than a benchmark comparison.

5.1 Persistence stability

Let X denote the fixed bin grid of a Stage-1 candidate domain, and let $f, g : X \rightarrow \mathbb{R}$ be two tame filtration functions on X – for instance, f built from the true underlying contact probability at each bin pair, and g built from an observed, noisy realization of the same matrix, using the same monotone transform in each case (Section 4.2, option (b), the cubical sublevel-set filtration). The stability theorem for persistence diagrams [5] states

$$d_B(\text{Dgm}(f), \text{Dgm}(g)) \leq \|f - g\|_\infty, \quad (1)$$

where d_B is the bottleneck distance between the two diagrams. A cubical filtration on a fixed grid is the natural choice here, rather than a Rips complex built from a distance matrix, because (1) applies directly to two functions on the same domain without an additional metric-distortion argument.

A standard consequence of bottleneck matching is that any feature in $\text{Dgm}(f)$ with persistence exceeding $2\|f - g\|_\infty$ cannot be matched to the diagonal by a bottleneck-optimal matching at distance $\|f - g\|_\infty$, and therefore has a corresponding feature in $\text{Dgm}(g)$ within the same distance. In words: a sufficiently persistent feature is guaranteed to survive the perturbation from f to g .

5.2 Bounding Hi-C count noise

Model the observed contact count at a bin pair as Poisson with mean equal to the true intensity, following the noise model underlying Fit-Hi-C-style statistical testing. For a bin pair with true contact probability p and effective local sequencing depth N , a Bernstein-type concentration bound gives, for any $\delta \in (0, 1)$,

$$\Pr \left[|\hat{p} - p| \leq \sqrt{\frac{2p \log(2/\delta)}{N}} + \frac{\log(2/\delta)}{3N} \right] \geq 1 - \delta. \quad (2)$$

Taking a union bound over the B bin pairs comprising the candidate domain (replacing δ with δ/B in (2)) gives a single sup-norm bound holding simultaneously across the domain, with probability at least $1 - \delta$:

$$\|f - g\|_\infty \leq \varepsilon(N, \delta, B) := \sqrt{\frac{2p_{\max} \log(2B/\delta)}{N}} + \frac{\log(2B/\delta)}{3N}. \quad (3)$$

5.3 The resulting guarantee

Combining (1) and (3):

Proposition. With probability at least $1 - \delta$, any candidate boundary in a domain of B bin pairs at local sequencing depth N whose persistence value exceeds $2\varepsilon(N, \delta, B)$ survives the perturbation from the true contact intensity to its noisy observation; that is, a corresponding feature of comparable persistence is guaranteed to be present in the diagram computed from the true, noise-free intensity.

We call a candidate boundary *ε -robust at confidence $1 - \delta$* when this condition holds, and report this as a calibrated confidence label alongside the raw persistence value, rather than replacing the empirical ranking comparison in Claim 1.

5.4 A necessary caveat, and the empirical alternative

Hi-C counts are not exactly Poisson in practice: biological replicate variability, GC-content and mappability bias, and distance-dependent decay all introduce overdispersion relative to a pure Poisson model, which is why Fit-Hi-C2 and HiC-DC+ use negative-binomial models instead. Two consequences follow, and both are part of the planned evaluation rather than an unresolved gap left for later. First, (2) can be replaced with an analogous negative-binomial concentration bound, widening ε to account for overdispersion; the argument structure in (3) and the Proposition above is unchanged. Second, and more decisively, we do not rely on the parametric bound alone: $\|f - g\|_\infty$, and hence a directly measured ε , can be estimated empirically by computing the bottleneck distance between persistence diagrams built from two biological replicates of the same region, with no distributional assumption at all. The parametric bound in (3) is used as a sanity check that this empirically measured ε falls in a reasonable range given the read depth; the empirical estimate, not the Poisson bound, is what Claim 1’s benchmark in Section 4.4 is anchored to.

6 Compute plan

The pipeline is implemented in Python (`cooler`, `cooltools` for Stage 1; `GUDHI` and `Ripser` for Stage 2), with genome-wide runs parallelized across chromosome, resolution, and cell line on

Modal, a serverless compute platform billed per CPU core-second. The workload is CPU-bound; no GPU compute is required. A single-chromosome, single-resolution prototype is run locally first to establish realistic wall-clock time and peak memory for Stage 2 before committing to a genome-wide parallel run, since the memory footprint of the sub-matrix persistent homology step, not the available compute budget, is the binding constraint on resolution choice.

The ultra-high-resolution variant in Section 4.3 is treated as a separate compute track with its own prototype, run locally on one chromosome at 500 bp before any genome-wide 200 bp run is attempted. The bin count per chromosome at 200 bp resolution is roughly two orders of magnitude larger than at 10 kb, so sparse matrix representations are not an optimization here but a requirement; dense-matrix assumptions anywhere in the Stage 1 code path would make a genome-wide run infeasible regardless of available Modal credit.

7 Limitations

We highlight five limitations directly, rather than leaving them for a reader to discover. First, topoHiC already establishes that whole-matrix persistent homology can identify TADs and loops, and we cannot benchmark against it directly due to the absence of public code; our comparison to it is necessarily indirect. Second, no ground-truth TAD set exists, so all benchmark claims here are claims about proxy-metric agreement, not claims about “true” TAD recovery. Third, the computational advantage of windowed, per-domain persistent homology over whole-matrix approaches is a claim about practical scalability at high resolution that must be demonstrated empirically with wall-clock and memory measurements, not assumed from the design alone. Fourth, the parametric confidence bound in Section 5 rests on a Poisson (or negative-binomial) noise model for Hi-C counts that is a simplification of true measurement noise; we treat the empirically calibrated ε , not the parametric bound, as the primary evidence for this reason, and report the parametric bound only as a sanity check. Fifth, Claim 3 concerns a resolution regime, 500 bp and 200 bp, where it is not yet established that any current method, including the one proposed here, recovers reliable boundaries at all; this is stated as an open empirical question rather than a claim of success, and a negative result confirming that current Hi-C data density is insufficient at these resolutions is treated as a legitimate outcome of the project, not a shortfall against it.

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